

Theorems:

Week 1: Theorem 1.1 Let A and B be statements. Then $A \Rightarrow B$ is equivalent to $\bar{A} \vee B$. That is, $(A \Rightarrow B) \Leftrightarrow (\bar{A} \vee B)$.

Theorem 1.2 Let A and B be statements. Then $A \Rightarrow B$ is equivalent to $\bar{B} \Rightarrow \bar{A}$. That is, $(A \Rightarrow B) \Leftrightarrow (\bar{B} \Rightarrow \bar{A})$.

Theorem 1.3 Let A and B be statements. Then $\overline{A \vee B}$ is equivalent to $\bar{A} \wedge \bar{B}$.

Theorem 2.4 Given any two real numbers $a < b$, there is some $r \in \mathbb{Q}$ such that $a < r < b$.

Theorem 2.5 The number $\sqrt{2}$ is irrational.

Week 2: